
1. Low Latency

- **Definition:** Minimizing the delay between a request and its response.
 - **Example:**
 - **Online Gaming:** Games like **Fortnite** or **Call of Duty** require low latency to ensure real-time responsiveness for players.
 - **Financial Trading:** Stock trading platforms like **Robinhood** or **Bloomberg** use low-latency systems to execute trades in milliseconds.
 - **Video Conferencing:** Tools like **Zoom** or **Microsoft Teams** rely on low latency to ensure smooth, real-time communication.
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2. High Throughput

- **Definition:** The ability to process a large volume of tasks or data in a given time.
 - **Example:**
 - **Streaming Platforms:** **Netflix** or **YouTube** handle high throughput by delivering video content to millions of users simultaneously.
 - **Database Systems:** **Apache Kafka** processes millions of messages per second for real-time data streaming.
 - **Web Servers:** **Google Search** handles billions of queries per day with high throughput.
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3. High Frequency

- **Definition:** Performing tasks or transactions at an extremely rapid rate.
 - **Example:**
 - **High-Frequency Trading (HFT):** Firms like **Citadel** or **Virtu** use algorithms to execute thousands of trades per second in stock markets.
 - **IoT Systems:** Smart home devices (e.g., **Amazon Alexa**) process high-frequency sensor data to respond to user commands.
 - **Scientific Computing:** Simulations like **weather forecasting** analyze high-frequency data from sensors and satellites.
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4. Fault-Tolerant

- **Definition:** Systems that continue operating even when components fail.
 - **Example:**
 - **Cloud Platforms:** **AWS (Amazon Web Services)** and **Microsoft Azure** use fault-tolerant architectures to ensure uptime even during hardware failures.
 - **Aerospace Systems:** Airplane control systems are designed to be fault-tolerant to prevent crashes in case of component failure.
 - **Databases:** **Google Spanner** is a fault-tolerant distributed database that ensures data availability even during outages.
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5. Multi-Tenant

- **Definition:** A system that serves multiple users or organizations (tenants) from a single instance.
 - **Example:**
 - **SaaS Platforms:** **Salesforce** serves multiple businesses on a single platform, isolating each tenant's data.
 - **Cloud Storage:** **Google Drive** or **Dropbox** allow multiple users to store and access files on shared infrastructure.
 - **Web Hosting:** Platforms like **WordPress.com** host multiple websites on a single server.
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6. Platform-Agnostic

- **Definition:** Software or systems that work across different platforms (e.g., operating systems, hardware).
 - **Example:**
 - **Web Browsers:** **Google Chrome** or **Firefox** run on Windows, macOS, Linux, and mobile devices.
 - **Programming Languages:** **Python** or **JavaScript** can run on various platforms without modification.
 - **Virtual Machines:** **Docker** containers allow applications to run consistently across different environments.
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7. Write-Once-Run-Anywhere (WORA)

- **Definition:** Code written once can run on any platform without modification.
 - **Example:**
 - **Java Applications:** Java programs run on any device with a **Java Virtual Machine (JVM)**, such as Windows, macOS, or Linux.
 - **Cross-Platform Frameworks:** **Flutter** or **React Native** allow developers to write mobile apps once and deploy them on both iOS and Android.
 - **Web Applications:** **Progressive Web Apps (PWAs)** work on any device with a web browser, regardless of the operating system.
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